ITT300 july 2023 - Exercise of final paper

Digital Electronic (MARA Technological University Perak)

UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION

COURSE	:	**INTRODUCTION TO DATA COMMUNICATION AND NETWORKING**
COURSE CODE	:	**ITT300**
EXAMINATION	:	**JULY 2023**
TIME	:	**3 HOURS**

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of three (3) parts: PART A (20 Questions)
   PART B (5 Questions)
   PART C (3 Questions)

2. Answer ALL questions from all three (3) parts:

   i. Answer PART A in the Objective Answer Sheet.
   ii. Answer PART B and PART C in the Answer Booklet. Start each answer on a new page.

3. Do not bring any material into the examination room unless permission is given by the invigilator.

4. Please check to make sure that this examination pack consists of:

   i. the Question Paper
   ii. an Answer Booklet – provided by the Faculty
   iii. an Objective Answer Sheet – provided by the Faculty

5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 11 printed pages

PART A (40 MARKS)

Please answer all questions.

QUESTION 1

In FDM with four input lines, there must be _______ guard bands.

A. 2
B. 3
C. 4
D. 5

(2 marks)

QUESTION 2

A multiplexer combines four channels using a time slot of 4 bits. What is the size of the frame?

A. 4 bits
B. 8 bits
C. 12 bits
D. 16 bits

(2 marks)

QUESTION 3

Which of the following is NOT TRUE about empty slots?

A. One of the input lines has no data to send
B. Empty slot occurs in statistical TDM
C. Input line has discontinuous data
D. None of the above

(2 marks)

QUESTION 4

Which of the following is the BEST technique to solve the problem for the bit rates of input lines that not in multiple integers?

A. Multilevel multiplexing
B. Multiple slot multiplexing
C. Pulse stuffing
D. One level multiplexing

(2 marks)

QUESTION 5

Which of the following statement is FALSE regarding controlled access method?

A. The secondary device can control the link after getting permission from primary device.

B. The data exchange is not executed if the destination is a primary device.

C. Secondary devices need to follow an instruction from primary device that controlled the link.

D. Data can be exchanged directly with secondary device bypassing the primary device.

(2 marks)

QUESTION 6

Choose a CORRECT statement regarding access methods in data communications.
i. One of the tasks in data link layer is to resolved access issues in shared media.
ii. In random access method, all stations have the same right to access the media.
iii. One station is assigned as supervisor to controlled other stations in random access methods.
iv. The sending stations need to consult each other in random access method to gain authorization to access the media.

A. i, ii, iii

B. i, ii, iv

C. i, iii, iv

D. ii, iii, iv

(2 marks)

QUESTION 7

In CSMA method, the sending device need to sense or listen to the media first before trying to use it to reduce a/an _________.

A. noise

B. error

C. collision

D. congestion

(2 marks)

QUESTION 8

In two-dimensional parity check, _____.

i. a block of bits is organized in a row
ii. it increases the likelihood of detecting burst errors
iii. it can detect all single-bit errors
iv. it can detect burst errors as long as the total number of bits changed is even

A. i and ii
B. ii and iii
C. i and iv
D. iii and iv

(2 marks)

QUESTION 9

Given a dataword 1001 and a divisor of 1101. What is its dividend?

A. 10010110
B. 1001110
C. 10010000
D. 1001000

(2 marks)

QUESTION 10

Below are the fields contained in the standard Ethernet frame EXCEPT ____.

A. cast
B. source address
C. preamble
D. data length or type

(2 marks)

QUESTION 11

10Base5 implementation uses a ____ topology and ____ cable.

A. bus; thick coaxial
B. bus; thin coaxial
C. star; twisted-pair
D. star; thick coaxial

(2 marks)

QUESTION 12

A multicast destination address _____.

A. means the frame comes from many stations.
B. defines only one recipient in one-to-many relationship.
C. defines a group of addreses in one-to-many relationship.
D. is a spacial case of multicast address and the recipents are all stations on the network.

(2 marks)

QUESTION 13

What is the function of Gigabit Medium-Independent Interface (GMII)?

A. It is the medium-independent tranceiver that perform encodes and decodes.
B. It defines how the reconciliation sublayer is to be connected to the PHY sublayer.
C. It connects the transceiver to the medium.
D. It sends parallel data to the PHY sublayer.

(2 marks)

QUESTION 14

"The network is divided into two or more networks. The capacity is shared between smaller network and the collision domain becomes smaller." These statements are refer to ____.

A. Bridge Ethernet
B. Switch Ethernet
C. Full-duplex Ethernet
D. Standard Ethernet

(2 marks)

QUESTION 15

1000Base-CX technology ____.

i. requires a bandwidth of 1.5GHz.
ii. uses 2-wires of shielded twisted-pair cable.
iii. is encoded using NRZ encoding
iv. has a maximum cable length of 25 meters.

A. i, ii and iii
B. i, ii and iv
C. ii, iii and iv
D. i, ii, iii and iv

(2 marks)

QUESTION 16

The number of addresses in class C block is

A. 65,534
B. 16,777,216
C. 512
D. 256

(2 marks)

QUESTION 17

The size of IPv6 address is _____ long.

A. 16 bytes
B. 14 bytes
C. 12 bytes
D. 10 bytes

(2 marks)

QUESTION 18

Which of the following statements is NOT true?

A. IP address is a logical address in the network layer of the TCP/IP protocol suite.
B. In classful addressing, the portion of the IP address that identifies the network is called the hostid.
C. Addresses in class D is used for multicasting.
D.. Subnetting increases the number of 1's in the mask.

(2 marks)

QUESTION 19

A subnet mask in class B has twenty 1s. How many subnets does it define?

A. 8
B. 16
C. 64
D. 128

(2 marks)

QUESTION 20

An organization is granted a block; one address is 2.2.2.64/20. The organization needs 10 subnets. What is the subnet prefix length?

A. /20

B. /24

C. /25

D. /28

(2 marks)

PART B (25 MARKS)

Please answer all questions.

QUESTION 1

Polling works with topologies in which one device is designated as a primary station and the other devices are secondary stations. Briefly explain (with diagram) the Select function in polling access method to coordinate access to the link.

(5 marks)

QUESTION 2

Differentiate between random access and controlled access. Give an example method available for each access.

(5 marks)

QUESTION 3

LANs provide benefits for users. List the benefits of using a networked over a stand-alone computer

(5 marks)

QUESTION 4

What are the maximum networks and hosts in a class A, B and C network?

(5 marks)

QUESTION 5

What is the number of network IDs in a Class C network?

(5 marks)

PART C (35 MARKS)

Please answer all questions.

QUESTION 1

a) Five (5) voice channels are to be multiplexed together with each channel occupying a 5 KHz bandwidth.

i) Draw the diagram if the multiplexing process uses guard bands of 500 Hz between each channel.

(3 marks)

ii) Calculate the total bandwidth of the link.

(1 mark)

iii) Give the reason, why guard bands are used in FDM?

(1 mark)

b) Three (3) channels, one with a bit rate of 50 kbps and others with a bit rate of 100 kbps, are to be multiplexed. If the multiple-slot allocation strategy has been applied:

i) Identify the frame rate?

(1 mark)

ii) Determine the frame size?

(1 mark)

iii) What is the frame duration?

(1 mark)

iv) Find the bit rate of the link and the bit duration?

(2 marks)

QUESTION 2

a) A dataword X^4+X^2 has successfully sent to a receiver without error. A common divisor of X^2+X^1+1 was applied. Convert the polynomial form into binary form and show the encoding and decoding process.

(10 marks)

b) Given a codeword formula, n=k+r, where k is the dataword and r is the check bit. If n=3, the possible number of generated codewords is 8. List all the possible bits of the dataword, k given the check bit is equal to 1, r=1.

(4 marks)

QUESTION 3

a) One of the usable IP address in BitNetwork is given as follows:

IP address: 195.190.10.10

Subnet mask: 255.255.255.0

i) Determine the class.

(1 mark)

ii) Determine the network address.

(1 mark)

iii) Determine the broadcast address.

(1 mark)

iv) Determine the number of usable IP address.

(2 marks)

b) BitNetwork is moving to new offices and the network administrator decided to create four subnets from the initial IP address (195.190.10.0) to accommodate the change. Show the working steps for each question.

i) Determine the address spaces for each subnet.

(1 mark)

ii) Determine the new subnet mask.

(1 mark)

iii) Determine the network address of the third subnet.

(1 mark)

iv) Determine the broadcast address of third subnet.

(1 mark)

v) Given an IP address 195.190.10.216 after the subnetting process.

(2 marks)

END OF QUESTION PAPER

